CIGS-TTVAE: Transcriptome Reconstruction from Target Gene Panel

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Introduction

- Single-cell RNA sequencing is needed for gene expression analysis, but computationally and financially expensive
- Only sequencing a panel of genes, as in the **Chemically-Induced Gene Signatures (CIGS)** dataset is cheaper

Objective

Train a model to reconstruct the expensive, full ~30,000 gene transcriptome from the ~3000 gene panel measured in CIGS

Methods

- 5,000 DMSO cell subset from **Tahoe100M** dataset, tokenized by Variational Gaussian Mixture Model
- **VAE Architecture:** FT-Transformer encoder with gene input layer and 2 transformer attention blocks mean pooling to 32-dim latent space, to standard MLP decoder
- **2 Model Iterations:** (1) MSE loss with KL annealing, (2) Weighted MSE loss with MMD regularization

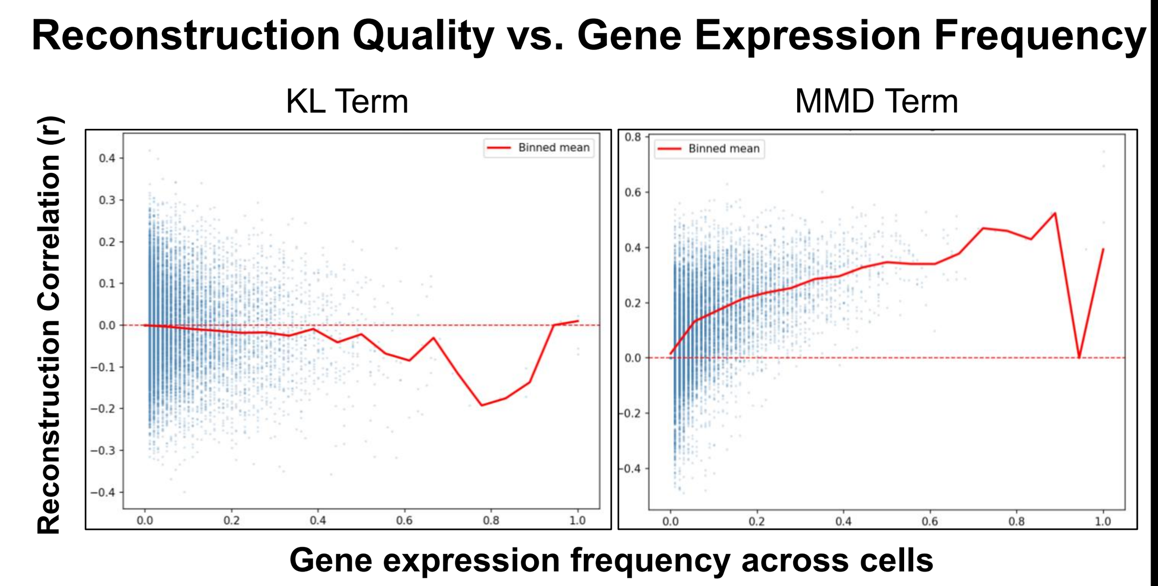
Results

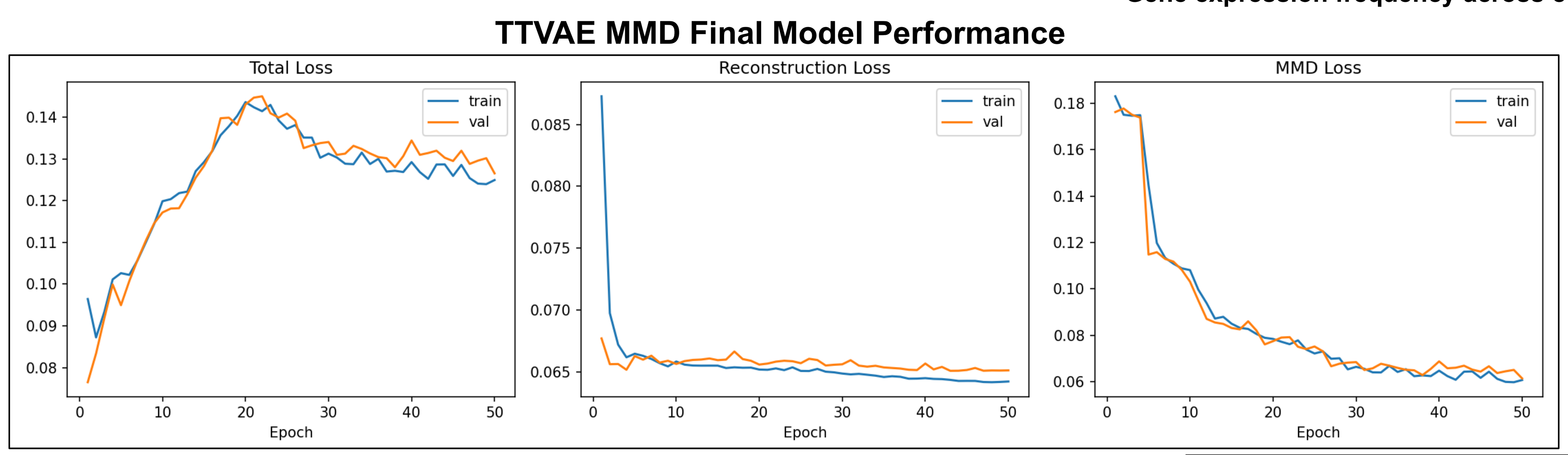
Results were very promising:

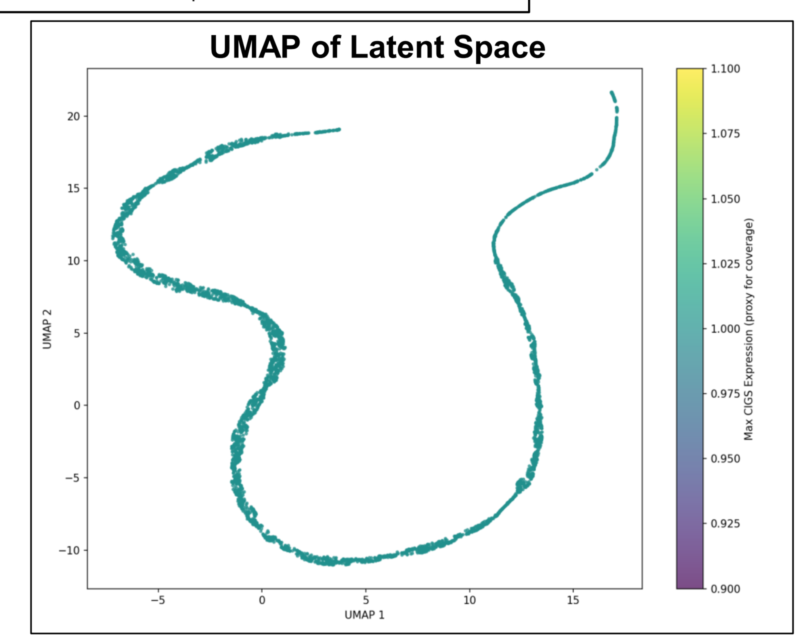
KL – RL: **0.02**, genes r > 0.3: **1040**

MMD – RL: **0.065**, genes r > 0.3: **1542**

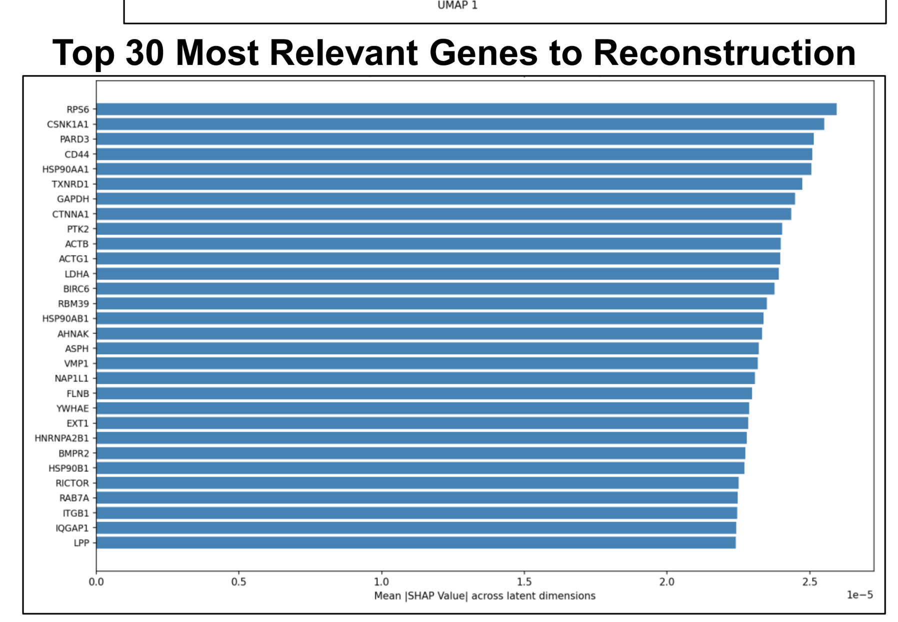
However, biological interpretability of KL was low due to collapse. MMD traded lower RL for a structured latent space.







UMAP of latent space shows a **continuous ribbon manifold**, consistent with cell cycle progression as the dominant axis of variation in proliferating DMSO control cells



SHAP analysis identified top CIGS drivers of latent representation: all three HSP90 members ranked top 30. Suggests protein folding and stress response are key sources of cell-to-cell variation

Discussion and Future Work

- Successfully learned a biologically structured latent space from unlabeled DMSO control cells, evidenced by SHAP
- This suggests meaningful reconstruction can be recovered from panel set of genes, though results still tied to gene frequency
- Larger training sets and deeper transformer architectures with additional attention blocks are needed to improve reconstruction of lowly expressed genes and capture finer cell-to-cell variation

References

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[3] J. Zhang et al. Tahoe-100M: A Giga-Scale Single-Cell Perturbation Atlas for Context-Dependent Gene Function and Cellular Modeling. bioRxiv, 2025.

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